

Group projects Programming in Python II

Deadline: 28/02/2025, 23:59, please submit your *.py or *.ipynb files using the designated folder on the K2 site of the course.

Project 1

The task is to predict house prices in different communities based on air pollution level (represented by the concentration of nitrogine oxid), distance from employment centers, average student-teacher ratio in local schools and average number of rooms, and crime rate The variables are as follows:

- *price*
median housing price
- *crime*
crimes committed per capita
- *nox*
nitrogine oxide, parts per 100 mill.
- *rooms*
avg number of rooms per house
- *dist*
weighted dist. to 5 employ centers
- *stratio*
average student-teacher ratio
- *lprice*
 $\log(\text{price})$
- *lnox*
 $\log(\text{nox})$

The .csv file can be found at
<https://kaggle.com/datasets/f06415afa98ac5a0a50ad7fdec3c4b526023dcf2a1b39435e6f2b6a43eddf88e> or downloaded from the assignment in K2 (hprice2_merged.csv)

Use various models (linear regression, SVM, decision tress, neural networks) for predicting house prices. Compare the various models in terms of accuracy of

prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting houses prices. Which variables give the best model for predicting house prices ? Try to predict the logarithm of house prices instead of house prices. Do you see any improvement in performance of the models ?